

MATH 619V

Topics in Algebra:

Elliptic Curves

Fall 2023

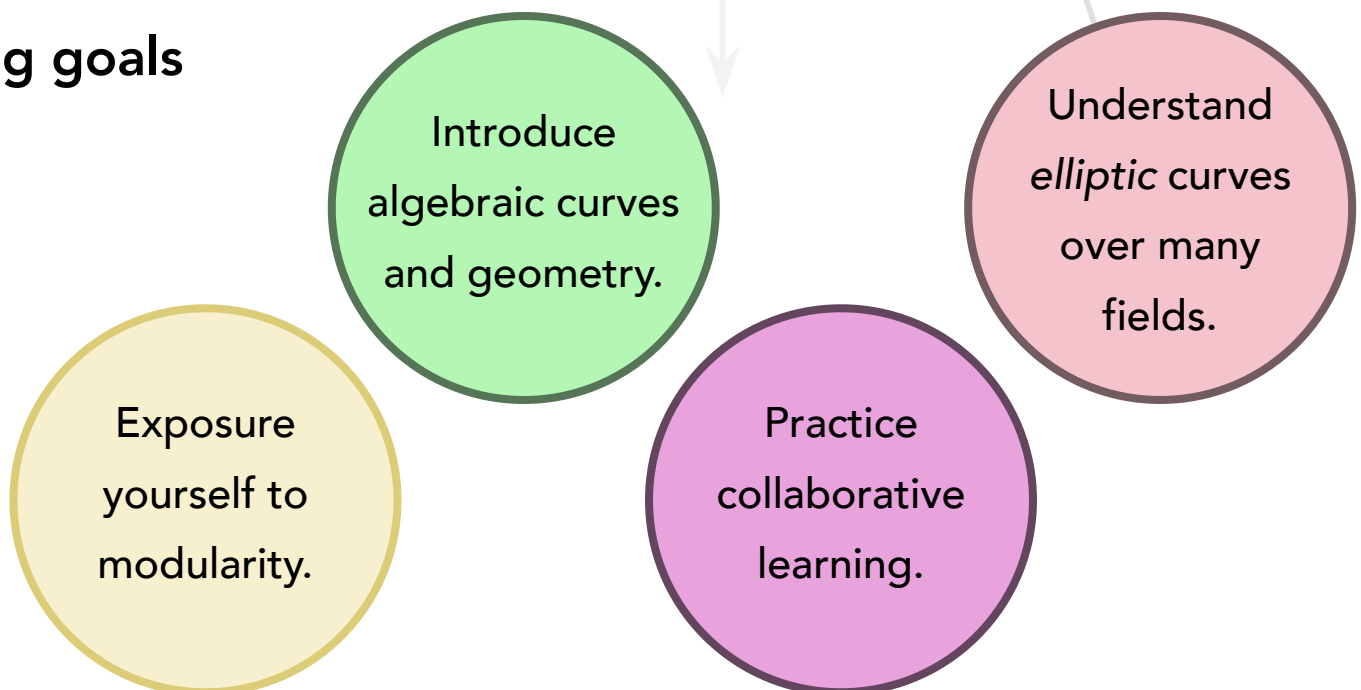
Professor John Bergdall
bergdall@uark.edu

MWF 11:50 AM - 12:40 PM
Classroom: SCEN 322

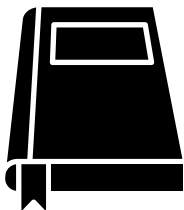
Overview

Elliptic curves belong to the intersection of algebraic geometry and number theory. They're *also* central in the 1994 resolution of Fermat's Last Theorem by Wiles and Taylor—Wiles. This seminar course marches through foundational theory toward a 10,000 foot view of that 30-year old breakthrough.

Learning goals



Basic resources



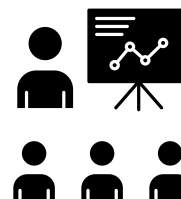
References

Elliptic curves ('06) by Milne
Algebraic curves ('69) by Fulton
(Free)



Course materials

References and all other
materials on Blackboard.
(Free)



Office hours (SCEN 220)

Monday	10:35-11:30
Wednesday	2:00-3:00
Thursday	2:00-3:00

Tasks

Course tasks are divided into *Supporting Tasks* and *Performance Tasks*. Supporting Tasks keep the course moving. Performance Tasks reflect your understanding. **Assignment and grade details on pages 3-5.**



Supporting Tasks

Reading Questions and Class
Contributions.



Performance Tasks

Homework Questions and
Mastery Assessments.

Turn-in

You will never turn-in work for this class. Some work will be completed on annotation software, while some occur in asynchronous discussion.

Revisions

You are expected to revise your work, to reinforce your learning.

- Worked Exercises may be continuously revised until the end of the semester.
- Learning Conversations may be revised through *written clarification* within one week.

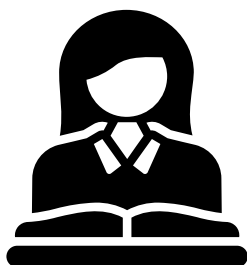
Late work

There are few deadlines in the course, so little opportunity to turn in late work. The only caveat is:

- Weekly Reading annotations may be entered by *Monday* provided at least three (3) others have submitted their annotations already.

Supporting Tasks

There are two main ways you'll support learning in our class. **Weekly Readings** will be your first encounter with new ideas. **Class Contributions** are how you impact the learning environment.



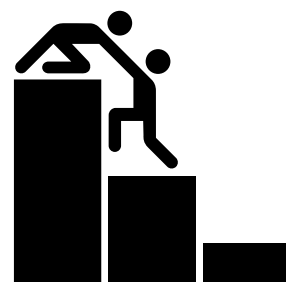
Weekly Readings

- **10-20 pages** *per week* from main references.
- **Annotations and comments** by Friday *morning*. There will be *no judgement* on the quality of your comments.
- **Frame class discussion** during the following week.

*We will fill collectively annotate an initial (short) reading on Wednesday August 23.

Class Contributions

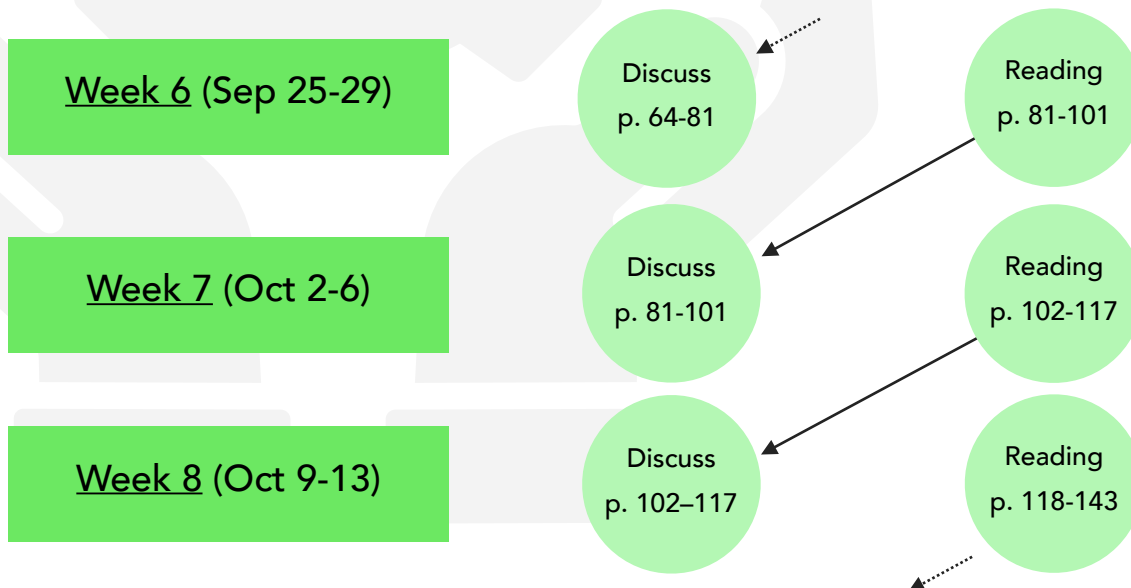
- **Discuss** on Blackboard.
- **Summarize** prior class.*
- **Reinforce** a friend's ideas.
- **Ask** questions in class.
- **Answer** questions in class.
- **Correct** my materials.



*On a rotating basis, students will summarize our prior discussion.

Example

We illustrate the chronology of Weekly Readings vs. Class Contributions in the following schematic (with made-up numbers).

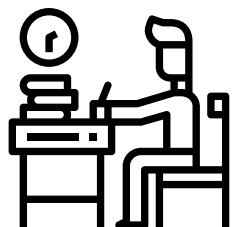


Supporting Task Grading

Supporting Tasks are counted per week. You **Earn a Week** if you:

- **Actively participate** in annotating the assigned reading.
- **Contribute at least once** to class, in a significant way.

Performance Tasks



Worked Exercises are opportunities to dig deeper and touch the theory. **Learning Conversations** ask you to perform the ritual of mathematical conversation, with your professor and colleagues.

Worked Exercises

- **2 Questions** solved asynchronously *most* weeks.
- **Lead Solvers** will *open* initial discussions.
- **Solving Team** will work *together* to craft solution.

*We will go through an example on Wednesday August 23.

Dates:

Oct 9 - Oct 13
Nov 13 - Nov 17
Dec 11 (Final)

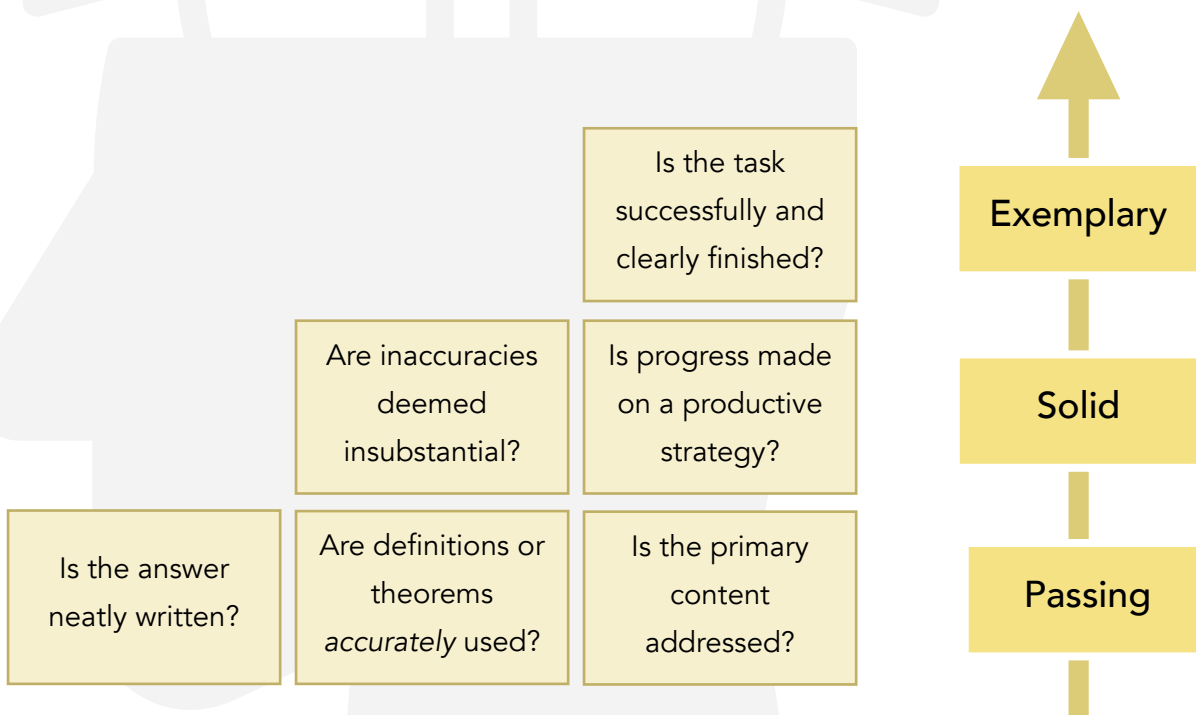
Learning Conversations

- **In-person conversations** three times in the semester.
- **Written clarifications** due within one week.
- **Progressively collaborative:** solo, paired, then entire group.



Performance Task Grading

Each Worked Exercise and Learning Conversation will be graded out of six points, based on qualities of *passing*, *solid*, and *exemplary* work.

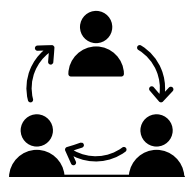


Final grade

Your final grade is determined by thresholds. We may make *more generous* thresholds later in the semester.

	To earn a C...	To earn a B...	To earn an A...
Supporting Tasks	Earn 9 Weeks	Earn 12 Weeks	Earn 14 Weeks
Performance Tasks	at least 50%	at least 70%	at least 90%

Tips for success



Collaborate!
With peers.



Post! To
Blackboard.



Ask! Me early
and often.

Tips for the stuck and frustrated.

- Take a break. Walk away.
- Explain to someone why you are stuck.
- Check hypotheses or assumptions.
- Work out a single example.
- Keep going!

Further opportunity

The **Arizona Winter School** (AWS, <https://swc-math.github.io>) is a yearly graduate student experience held at the University of Arizona. Those pursuing research in algebra, number theory, and algebraic geometry are advised to attend as frequently as possible. For the 2024, the topic is **abelian varieties**. As preparation, the Preliminary Program (PAWS, so cute) features two six-week courses on *elliptic curves* and *abelian varieties*.

A student that completes this course is in good position to participate in the 2024 AWS. After this course, they should study the PAWS material and further learn basic algebraic geometry. I will help pursue any and all preparation like this. Actively tell me *early* if you are interested.

Further Resources

Appointments

Office hours will not always fit your schedule. Email me to set appointments. In your initial message, please explain (i) what you want to talk about, (ii) *your* availability and (iii) if meeting in person is important (Zoom is possible).



Email

Your questions are important. Email me or post to Blackboard. (I may ask you to do this anyways.) Responses may take 24 hours, but send a reminder if I do not reply.

Access

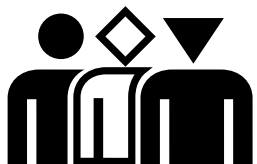
I want you to pass this class, with success. If you find resources not being provided, tell me and I will work with you to fix the issue. For some students, the Center for Educational Access (cea.uark.edu) can coordinate student accommodation requests. Please request your accommodation letter early semester and meet me to discuss it.



Class formation

Names and pronouns

You deserve to be called how you want. Let me know your preferred name and pronouns any time. When meeting one another, re-introduce yourself *frequently*. Everyone's names, pronouns, etc. should be known and properly used.



Diversity

I pledge attention to your identity and experience, regardless of age, background, beliefs, ethnicity, gender identity and expression, national origin, racial identity, religious beliefs, sexual orientation, and any other visible or non-visible categories. You are expected to do the same for myself and your peers.

Academic integrity

I trust my students to follow (i) the University of Arkansas Academic Integrity Policy and (ii) my own explicit assignment instructions. When you have questions about a boundary or rule, ask me rather than assuming you know the answer.



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